Poverty amid Progress:
The Hidden Cost of a Cold Shock in Mongolia

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The 2023/24 dzud killed **nine million livestock, one in seven.** Yet **poverty kept falling fast.**

A dzud is an extreme winter of deep cold, heavy snow and ice-locked pasture. Outside the capital, one household in three herds.

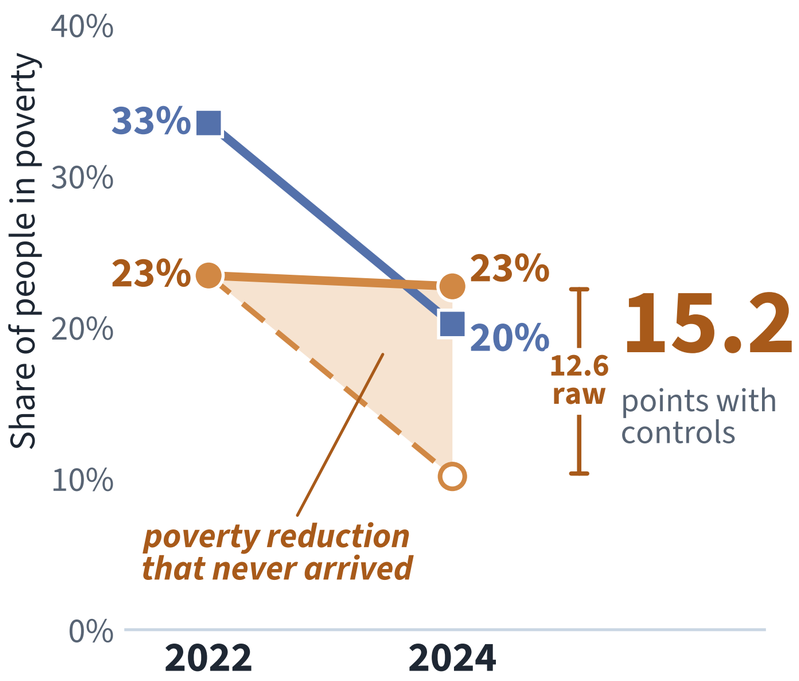
What did falling poverty hide? For whom, and how?
A large, concentrated loss. Herders most, but not alone.

Severely affected districts · Less affected districts · Severely affected, had poverty fallen like the rest

Underneath the decline

All households

Flat where the winter hit hardest, falling elsewhere



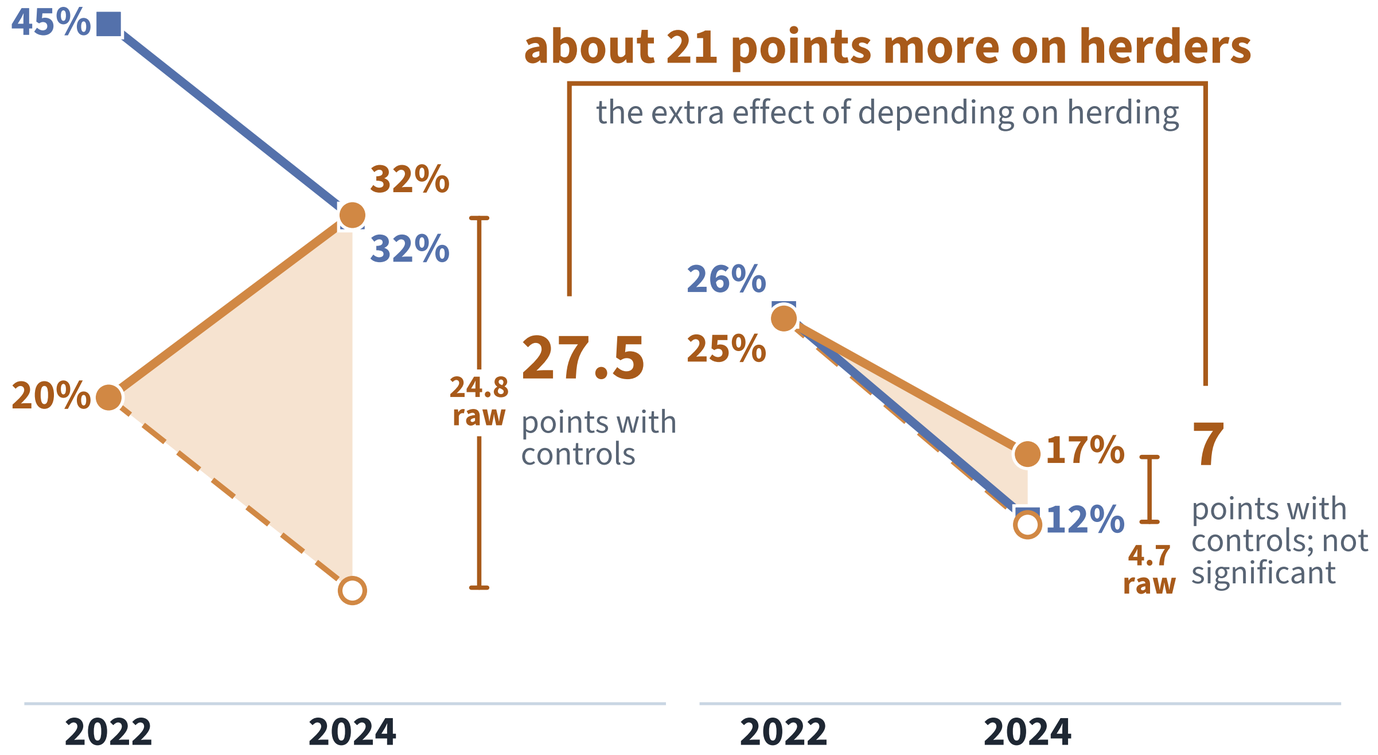
Who carried it: herders against non-herders

Herders

Rising where the winter hit hardest

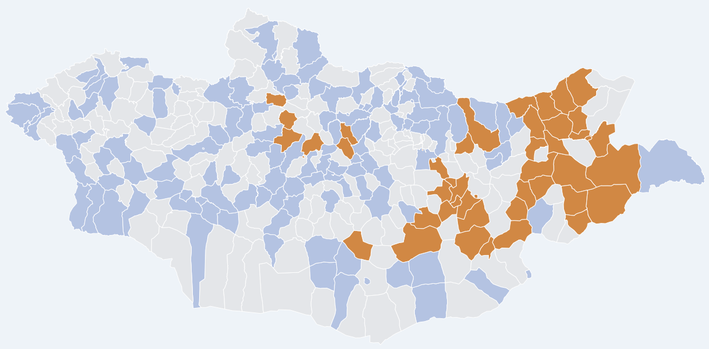
Non-herders

Falling less, not significantly



Same households, 2022 and 2024, outside the capital. Dashed: poverty in the severely affected districts had it fallen like the rest. Bracket: the raw gap. Large number: the estimate with controls.

1 The study design



Severely affected: 46 districts whose cold or snow, against their own history, was in the top 15 percent. Lost **27 percent** of livestock.

Less affected: 129 others. Lost **9 percent**.

Difference-in-differences: the poverty change in **orange** minus **blue** districts, same **2,128 households**.

Triple difference: that difference for herders minus non-herders. Anything that hit both equally, such as blocked roads, cancels out.

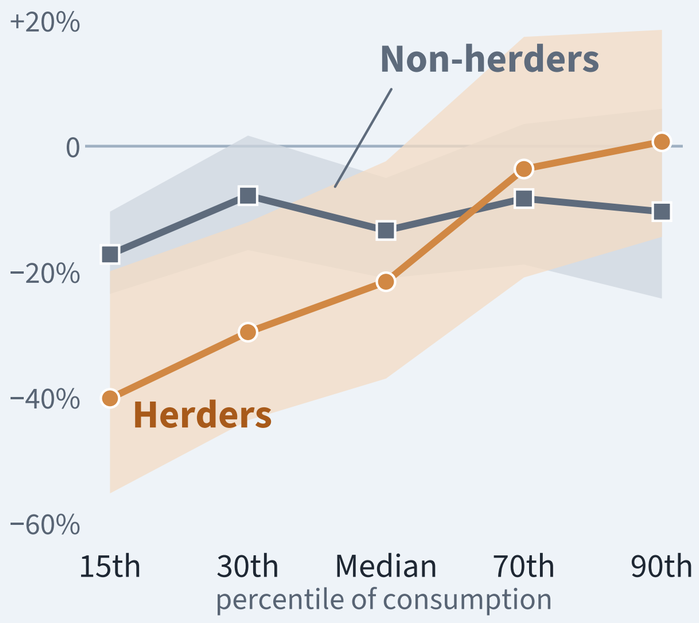


- ✔ Parallel before the winter
- ✔ Placebo winter (2018/19): no effect
- ✔ At least 9.4 points if poorer districts catch up
- ✔ Less affected were also hit: if anything, an understatement

2 For whom, and how

Where the loss landed

Herders crossed the poverty line; poor non-herders fell further below it.



Consumption against the less affected districts, by percentile. Bands: 90 percent confidence intervals.

Herders: deaths, not distress sales

Herds shrank **36 percent**, against 13 percent elsewhere.

Non-herders: the food channel

Calories about **12 percent** lower. They had less of their own food as prices rose, and food took 47 percent of poor budgets.

3 Policy implications

What the winter cost

32,000 to 52,000 people pushed into or kept in poverty; **$65 to 88 million** of consumption lost in 2024.

Reach beyond herders

Poor non-herders lost food, not herds. Relief aimed at herders misses them.

Open question: can early aid help?

After: emergency response — **$15m**

Before: $238 per herder household — **$9m**

The $9m assumes we know which districts will be hit, but December's forecast flagged most of the country. In the trial, small herders kept more animals; on average, there was no effect.

A falling national headcount is not evidence that a shock was absorbed.

Extreme winters will come again.

When the numbers show progress, look for the hidden cost.

Poverty rates are the authors' calculations. Data: HSES 2022 and 2024 household panel (National Statistics Office of Mongolia); CRU TS 4.09 temperature; MODIS snow cover; NSO livestock statistics. Grey on the map: no panel households, or the capital. Emergency response: UNDP (2025). Pre-winter transfer: Mogge, Roeckert and Kraehnert (2025, JAERE). Frost texture generated with ChatGPT. The views expressed are the authors' and do not necessarily represent those of the World Bank, its Executive Directors, or the governments they represent.